COMP396 Weekly Meeting Minutes

(Week 5)

Date & Time: Friday, February 27th, 2026, 14:00–18:00

Location: Harold Cohen Library, Upper Ground Floor, Group Study Table 2

Note Taker: Wong Haw (WH)

Attendees: Zhang Olivia (ZO), Zhao Saria (ZS), Xue Ella (XE), Wong Haw (WH)

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1. Agenda

- Review the completion status of the strategy development plan and the initial draft of the Design Document.
- Evaluate the refined logic for asset classification and strategy mapping based on performance metrics.
- Summarize the progress of ZS's Python implementation within the unified TeamStrategy class.
- Allocate specific report-writing and technical tasks for the upcoming week.

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2. Discussion & Decisions

2.1. Task Review and Result Confirmation

We successfully completed all tasks assigned during the previous meeting:

- **Report Preparation**: ZO, WH, and XE collaboratively completed Section 01 of the Design Document, "Final Selection of Strategy", including the overall strategic framework and the foundational logic for the three sub-strategies.
- **Code Development**: ZS finished the core framework and specific indicator logic for the three sub-strategies within the single-class architecture.

2.2. Refined Asset Classification Rationale (Provided by ZS)

ZS presented a detailed justification for asset selection, building upon the OLS/ADF structural analysis conducted in Week 4 :

- **Primary Metric**: ZS decided to adopt the **Return/MaxDD Ratio** as the core classification indicator.
- **Justification**: ZS argued that this ratio simultaneously accounts for profitability and downside risk, noting that Maximum Drawdown is a more realistic measure of the losses sustained during actual trading compared to standard volatility.
- **Classification Rules**:
    - **Trend/Momentum Assets**: Ratio >= 1.1. ZS considers this level of return sufficient to compensate for drawdown risk. Assets identified: **Series 01, 03, 04, 06, 09**.
    - **Mean-Reversion Assets**: Ratio < 1.1 and Positive Return. ZS noted these assets show lower returns or excessive drawdowns, making them unsuitable for trend-following. Assets identified: **Series 02, 07, 08, 10**.
    - **Short-Selling Assets**: Negative Return. Asset identified: **Series 05**.
- **Threshold Optimization**: ZS explained that the **1.1** threshold was finalized after iterative testing; a threshold of 1.0 would misclassify the mean-reverting Series 08 into the trend group, while 1.2 would exclude the clearly trending Series 09.
- **Outputs**: ZS generated the research/figures/asset_classification.csv and a corresponding classification scatter plot.

2.3. Strategy Implementation Progress (Summary of ZS's Code)

ZS demonstrated the development progress of the TeamStrategy class using the backtrader framework:

- **Architecture**: ZS implemented a unified class inheriting from bt.Strategy, utilizing a params

dictionary to configure capital allocation (70% for Trend assets, 30% for others).

- **Asset Mapping**: ZS defined a comprehensive asset_params dictionary to match specific sub-strategies to each series:
    - **Trend**: Breakout (S1), Above-MA (S3, S6, S9), and MA Crossover (S4).
    - **Mean-Reversion**: Z-Score (S2, S10), Bollinger Bands (S7), and RSI (S8).
    - **Short**: RSI-based logic for Series 05.
- **Indicator Management**: ZS dynamically creates and stores technical indicators (SMA, RSI, Highest, Lowest) in a nested dictionary for per-asset reference.
- **Weight Control**: ZS developed get_portfolio_value() and get_target_value() helper functions to ensure per-asset sizing adheres to our 14% (Trend) and 6% (Other) weight limits.

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3. Task Allocation

ID	Task Description	Owner	Due Date
D1	Complete the report section: "Detailed Logic of the Three Sub-strategies".	ZO	Mar 06
D2	Complete the report section: "Capital Allocation Scheme".	WH	Mar 06
D3	Assist ZO and WH in revising and formatting tables for the first two sections of the report.	XE	Mar 06
D4	ZS is responsible for continuing the code development, ensuring that all parameters reach the highest possible scoring points (including the presentation of True PD Ratio, Activity, etc.).	ZS	Mar 06

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4. Current Challenges and Issues

- **Performance Metric Verification**: ZS noted the need to ensure the **PD Ratio** calculated within our code precisely matches the backtester's internal logic to avoid scoring discrepancies during the CA2 evaluation.

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5. Next Meeting Arrangement

- **Date & Time**: Friday, March 6th, 2026, 14:00–16:00
- **Location**: Harold Cohen Library, Upper Ground Floor, Group Study Table 2
- **Key Agenda**:
  - Finalize the sub-strategy logic and capital allocation sections of the report.
  - Conduct a code review of ZS's unified T-Strategy file.
  - Verify preliminary performance indicators (PD Ratio and Activity) on Part 2 data.